

Rational-Flux Anisotropic Harper Dynamics: Chambers Phase Collapse and the Algebraic Bloch-Band Atlas

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Abstract

For every reduced rational magnetic flux and every positive hopping anisotropy, we fix one magnetic-cell phase convention and derive the exact Chambers characteristic identity. The proof exposes why the vertical transfer trace has only three Laurent modes and fixes both phase signs. We then recover the full two-dimensional spectrum as a polynomial preimage, give the exact algebraic band-contact criterion, prove Aubry duality, flux reversal and parity, and isolate the forced central contact at every even denominator. The one- and two-site cells are computed separately. An exact cyclotomic lane and 74,880 independently reconstructed Bloch fibers audit the result. Harper–Chambers theory is established background, and the final Route-A verdict remains negative despite natural quantization.

1 Magnetic owner and phase convention

Let $(p, q) = 1$, $q \geq 3$, and $\lambda > 0$. On $\ell^2(\mathbb{Z}^2)$ consider the bounded self-adjoint operator

$$\begin{aligned} (\mathbf{H}_{p/q, \lambda} \Psi)_{m, n} &= \Psi_{m+1, n} + \Psi_{m-1, n} \\ &\quad + \lambda e^{2\pi i p m / q} \Psi_{m, n+1} + \lambda e^{-2\pi i p m / q} \Psi_{m, n-1}. \end{aligned} \quad (1)$$

Fourier transformation in the second coordinate and magnetic Bloch reduction give the q -dimensional fiber

$$(\mathbf{H}(k_x, k_y)u)_m = u_{m+1} + u_{m-1} + 2\lambda \cos\left(k_y + \frac{2\pi p m}{q}\right) u_m, \quad u_{m+q} = e^{iqk_x} u_m. \quad (2)$$

Thus $X = qk_x$ is the *total* horizontal boundary phase. This distinction is essential: throughout,

$$D_{p/q, \lambda}(E; k_x, k_y) := \det(EI_q - \mathbf{H}_{p/q, \lambda}(k_x, k_y)). \quad (3)$$

This manuscript's first revision is the **Chambers phase-collapse owner**.

2 The Chambers polynomial

Put $\zeta = e^{2\pi i / q}$, $y = e^{ik_y}$, and

$$A_m(E, y) = \begin{pmatrix} E - \lambda(y\zeta^{pm} + y^{-1}\zeta^{-pm}) & -1 \\ 1 & 0 \end{pmatrix}, \quad M_q = A_{q-1} \cdots A_0. \quad (4)$$

Theorem 1 (Anisotropic Chambers identity). *There is a unique real monic polynomial $\mathcal{P}_{p/q, \lambda}$ of degree q such that*

$$D_{p/q, \lambda}(E; k_x, k_y) = \mathcal{P}_{p/q, \lambda}(E) - 2\cos(qk_x) - 2\lambda^q \cos(qk_y) \quad (5)$$

for every real E, k_x, k_y .

Proof. Since $\det A_m = 1$, the Floquet multipliers of M_q are $e^{\pm iqk_x}$. Both sides below are monic of degree q and have the same roots, hence

$$D_{p/q, \lambda} = \text{tr } M_q - 2\cos(qk_x). \quad (6)$$

Replacing y by $\zeta^p y$ cyclically permutes the potential sequence in (4). The monodromy changes by cyclic conjugacy, so its trace is invariant. Because p is invertible modulo q , a Laurent polynomial of y with degrees between $-q$ and q and this invariance can use only the modes $-q, 0, q$.

The coefficient of y^q has one contribution: select $-\lambda y \zeta^{pm}$ in every upper-left entry. It equals

$$(-\lambda)^q \zeta^{pq(q-1)/2} = -\lambda^q. \quad (7)$$

Indeed, odd q makes $q-1$ even, while even q forces p odd. Complex conjugation gives the same coefficient for y^{-q} . The remaining mode is independent of both phases, real, and monic; call it $\mathcal{P}_{p/q,\lambda}(E)$. Equations (6) and (7) prove (5). Monicity gives uniqueness. $\square$

The theorem is convention-complete: using the boundary character X in place of k_x rewrites the horizontal term as $-2 \cos X$, not $-2 \cos(qX)$. This is precisely the convention tested by the executable fibers.

3 Spectrum and algebraic band contacts

This revision is the **spectrum, duality, and edge owner**. Write

$$\mathcal{C}_{q,\lambda} = 2(1 + \lambda^q), \quad B_{p/q,\lambda}(E) = \mathcal{P}_{p/q,\lambda}(E)^2 - \mathcal{C}_{q,\lambda}^2. \quad (8)$$

Lemma 1 (Sourced cyclic-continuant evaluation). *If q is even, then*

$$\mathcal{P}_{p/q,\lambda}(0) = 2(-1)^{q/2}(1 + \lambda^q). \quad (9)$$

Proof. Lamoureux and Mingo use $h_{\theta,L} = u + u^{-1} + (L/2)(v + v^{-1})$. Their Theorem 2.5 proves the vanishing of every intermediate cyclic-matching sum, and their Corollary 2.6 gives, for even q ,

$$\Delta_{p/q,L}(0) = 2(-1)^{q/2}(1 + (L/2)^q) \quad [5].$$

Our diagonal potential is $2\lambda \cos(\cdot)$, so their parameter is $L = 2\lambda$. Comparing the phase-independent monic term in their Chambers formula with Theorem 1 identifies $\Delta_{p/q,2\lambda} = \mathcal{P}_{p/q,\lambda}$: their defining vertical phase $\pi/(2q)$ kills the explicit q th cosine, while our transfer trace then equals $\mathcal{P}$. This also fixes our $\det(EI - H)$ convention. Substitution gives (9). Thus this all- q step is sourced rather than inferred from the finite cyclotomic audit. $\square$

Theorem 2 (Complete Bloch spectral atlas). *The full two-dimensional spectrum of (1) is*

$$\text{Spec}(H_{p/q,\lambda}) = \{E \in \mathbb{R} : |\mathcal{P}_{p/q,\lambda}(E)| \leq \mathcal{C}_{q,\lambda}\}. \quad (10)$$

The two edge factors are actual fiber characteristic polynomials:

$$\mathcal{P}(E) - \mathcal{C}_{q,\lambda} = D(E; 0, 0), \quad \mathcal{P}(E) + \mathcal{C}_{q,\lambda} = D(E; \pi/q, \pi/q). \quad (11)$$

Consequently the algebraic band-edge multiset is the real zero multiset of B . An edge label is multiple precisely when

$$\mathcal{P}(E) = \pm \mathcal{C}_{q,\lambda}, \quad \mathcal{P}'(E) = 0. \quad (12)$$

Moreover,

$$\mathcal{P}_{p/q,\lambda}(E) = \lambda^q \mathcal{P}_{p/q,1/\lambda}(E/\lambda), \quad (13)$$

$$\mathcal{P}_{p/q,\lambda}(E) = \mathcal{P}_{(q-p)/q,\lambda}(E), \quad (14)$$

$$\mathcal{P}_{p/q,\lambda}(-E) = (-1)^q \mathcal{P}_{p/q,\lambda}(E). \quad (15)$$

If q is even, then

$$\mathcal{P}_{p/q,\lambda}(0) = 2(-1)^{q/2}(1 + \lambda^q), \quad \mathcal{P}'_{p/q,\lambda}(0) = 0, \quad (16)$$

so zero is a multiple central edge.

Proof. Magnetic Bloch theory is a direct-integral decomposition into (2). By (5), E lies in some fiber spectrum exactly when

$$\mathcal{P}(E) = 2 \cos(qk_x) + 2\lambda^q \cos(qk_y).$$

The phases vary independently, and their sum fills $[-\mathcal{C}_{q,\lambda}, \mathcal{C}_{q,\lambda}]$, proving (10). Putting $(k_x, k_y) = (0, 0)$ and $(\pi/q, \pi/q)$ in (5) proves (11). Both endpoint fibers are real symmetric: their total horizontal boundary phases are 0 and π , and their diagonal potentials are real. Thus both edge factors have only real roots, with multiplicity, and $B = (\mathcal{P} - \mathcal{C})(\mathcal{P} + \mathcal{C})$

is real-rooted. Equation (12) follows from $B' = 2\mathcal{P}\mathcal{P}'$ and $\mathcal{P} \neq 0$ at an edge. It is an algebraic coalescence criterion, not a claim that every remaining gap is open.

Exchange of the two lattice axes reverses the magnetic orientation and swaps the hopping amplitudes. After division by λ , magnetic Fourier duality identifies the result with $\lambda H_{(q-p)/q, 1/\lambda}$. Complex conjugation identifies the fluxes p/q and $(q-p)/q$. Comparing the phase-independent monic terms in (5) proves (13)–(14).

The bipartite involution $\Gamma\Psi_{m,n} = (-1)^{m+n}\Psi_{m,n}$ conjugates H to $-H$ and shifts both quasimomenta by π . Substitution in (5) proves (15). For even q , Lemma 1 gives the first formula in (16). Equation (15) makes $\mathcal{P}$ even, so $\mathcal{P}'(0) = 0$. This proves the derivative formula without extrapolating the finite exact lane. $\square$

Proposition 1 (Accumulated small-cell boundaries). *When coincident wrap neighbors are added rather than overwritten, Theorem 1 has the direct boundary polynomials*

$$q = 1, p = 0 : \quad \mathcal{P}(E) = E, \quad q = 2, p = 1 : \quad \mathcal{P}(E) = E^2 - 2(1 + \lambda^2). \quad (17)$$

The two $q = 2$ folded bands meet at zero. As $\lambda \downarrow 0$, the spectrum tends to the decoupled horizontal-chain interval $[-2, 2]$; reciprocal duality is not evaluated at $\lambda = 0$.

Proof. For $q = 1$ the scalar fiber is $2\cos k_x + 2\lambda\cos k_y$. For $q = 2$, write $X = e^{2ik_x}$ and $Y = e^{ik_y}$. The off-diagonal entries are $1 + X^{-1}$ and $1 + X$, while the diagonal entries are $\pm\lambda(Y + Y^{-1})$. Its determinant is

$$E^2 - 2(1 + \lambda^2) - (X + X^{-1}) - \lambda^2(Y^2 + Y^{-2}),$$

which proves (17) and the asserted contact. $\square$

4 Independent evidence

This revision is the **evidence, collision, and route owner**. The evidence exhausts all reduced fractions with $3 \leq q \leq 16$ and the five anisotropies $1/2, 2/3, 1, 3/2, 2$.

receipt	count
reduced fluxes / parameter panels	78 / 390
Bloch total-phase grid per panel	12×16
Hermitian fibers / eigenvalues	74,880 / 825,600
characteristic-determinant probes	224,640
exact cyclotomic reduced fluxes, $q \leq 10$	30

Table 1: Finite regression scale. Sampling is not the proof.

The producer obtains $\mathcal{P}$ from transfer-polynomial multiplication. The checker never imports it: it reconstructs $\mathcal{P}$ from the characteristic polynomial of one Hermitian reference fiber and then rebuilds every matrix in Table 1. A third lane works exactly in $\mathbb{Q}[\zeta_q]/(\Phi_q)$ with a symbolic anisotropy degree; it checks the Laurent support, both extreme coefficients, duality, reversal, parity, and (16) through $q = 10$. It is a regression check, not the all-denominator proof supplied by Lemma 1. Isolated replay, repaired-hash mutations and strict JSON/YAML parsers close the computational boundary.

5 Source and collision audit

Harper’s magnetic difference equation, Chambers’ rational-flux relation, and Hofstadter’s magnetic bands are classical [1, 2, 3, 4, 6]. Lamoureux and Mingo directly precede the cyclic-matching cancellation and even-denominator constant-term formula used in Lemma 1 [5]. We claim no priority for these objects or formulas. The package contribution is a convention-locked reconstruction, the explicit real endpoint-fiber edge factorization (11), and an independently replayable atlas.

The direct workspace neighbor is HCS-C15/HEN-O30. It owns a critical Weyl–Harper block $U + U^* + V + V^*$ at flux $1/3^m$ and the return of its top spectral edge along a Heisenberg congruence tower. It does not own the two-phase identity (5), all reduced fluxes, anisotropic duality, or the edge polynomial (8). C293 is a magnetic Grushin cylinder, C340 is a Lamé Floquet operator, and C356 is a QWZ Chern pump. No Chern, transport, edge-state, irrational-flux Cantor-spectrum, or Ten-Martini result is imported here.

6 Route decision and nonclaims

Reduced rational flux gives exact cyclotomic magnetic translations, earning $A0_{\text{WEAK_ARITHMETIC_RELATION}}$. It does not give rational-prime ownership, prime-power repetition, an isolated primitive-orbit ledger, or a logarithmic prime clock, so $A1_{\text{FAIL}}$. The finite source characteristic polynomial is neither a target Euler product nor a target Fredholm determinant, so $A2_{\text{FAIL}}$; no target continuation, functional equation, divisor or zero-counting law exists, so $A3_{\text{FAIL}}$. Equation (1) is already a bounded self-adjoint Hamiltonian, hence $A4_{\text{NATURAL_QUANTIZATION}}$, but its Bloch bands are not target zeros or a Hilbert–Pólya realization.

The overall verdict is ROUTE_A_REJECTED under the literal scope $\text{NO_BAD_EULER_OR_ROOT_NUMBER}$. No target arithmetic local datum, Euler factor, root number, automorphy, target-zero match, or Hilbert–Pólya operator is asserted. Route B remains locked.

References

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